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Ensemble Learning – Research Project

Project 1: results report

Non-linear and Interaction Features for Deep Forest

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Abstract

We report the results of the experiment in our final plan [1]: Deep Forest with columns added between layers, compared with the same cascade without them on 16 classification and 16 regression datasets (5,280 runs). None of the five hypotheses is supported. Every variant changes the median error by less than 1%, and for H1, H3, H4 and H5 the 95% intervals make gains above about 0.7% unlikely; the sensitivity checks agree. Products chosen by the layer error lose less than products chosen by the target (20 wins, 9 losses, not significant after Holm's correction), and no choice of products beats adding none. On two synthetic datasets with a planted product of mean-centred features the products cut the error by about 40%. The analyses suggested by the instructor find no gain either; they suggest (exploratory) that monotone activations, which nearly copy the predictions, make class probabilities worse. Exploratory follow-up runs suggest why products do not help: they carry information, but less than the extra columns cost, and on 6 of the 7 datasets where they help at the first layer, the cascade's own later layers gain more without them. Products of uncentred features do no better on the real datasets. A planted product that explains 6% of the target's variance is already enough for the products to help.

1 Introduction

The plan [1] proposed adding columns between the layers of Deep Forest [2] and compared each variant with the *base cascade*, which adds none. H1 and H2 concern feature–feature (XX) products chosen by correlation with the previous layer's error (*residual-XX*), compared with the base cascade (H1) and with random and target-chosen products (H2a, H2b). H3 concerns feature–prediction (XP) products chosen by mutual information, H4 the uncertainty and disagreement of the predictions, and H5 monotone activations of the predictions. We do not repeat the method, data or protocol. Analyses added after we saw the main results are exploratory and decide no hypothesis.

2 Experiments as run

Implementation and cost. We reimplemented the DF21 cascade [3] in scikit-learn with the package's defaults. With the same folds, it matches the package on two regression sets (RMSE within 0.3%) and is 0.5 and 2.9 points lower in balanced accuracy on Spambase and Statlog Vehicle, lower on 4 of 5 folds in both (Table A11). The planned differences may explain this: OOF predictions from fold models trained on two thirds of the rows, and balanced accuracy in the depth rule, which stops earlier here. Code and reproduction steps are in Appendix C. The main experiment (11 methods, 32 datasets, five folds × three seeds) gave 5,280 runs without failures in 38 CPU-hours, a third to a half of the plan's estimate; the sensitivity checks (one seed, 3,840 runs) took 59. Per run, the XX variants cost 1.17 times as much as the base cascade (median), XP-MI 1.46 times, and the non-linear and monotone columns 1.01–1.04 times. Nine data choices the plan leaves open (Table A10) and LightGBM's early stopping (patience 50, default metric) are ours. One deviation was found only after the runs: three datasets keep categories stored as numbers as numeric columns (Credit Default: sex, education, marital status; Auto MPG: origin; Energy Efficiency: orientation, glazing-area distribution), although the plan one-hot encodes categorical features. One-hot encoding them changes no conclusion (exploratory, Appendix B).

Table 1: The six comparisons that decide H1–H5 (32 datasets). W/L/T: datasets where the first method's mean error is lower, higher or equal. Median and Hodges–Lehmann estimate (a pseudo-median) of the relative error reduction (%; positive favours the first method), with the exact 95% interval; ties are kept. p: two-sided Wilcoxon signed-rank test; Holm: adjusted over the six. Support requires Holm $p < 0.05$ and a positive median.

	Comparison	W/L/T	Median	HL	95% CI	p	Holm p
H1	Residual-XX vs base	14/18/0	−0.66	−1.13	[−3.12, 0.12]	0.095	0.47
H2a	Residual-XX vs random-XX	16/16/0	0.02	0.13	[−0.99, 1.16]	0.82	1.00
H2b	Residual-XX vs target-XX	20/9/3	0.29	0.43	[0.06, 1.37]	0.020	0.12
H3	XP-MI vs base	9/17/6	−0.06	−0.20	[−0.93, 0.12]	0.27	0.81
H4	Non-linear vs base	14/12/6	0.00	0.00	[−0.72, 0.36]	0.89	1.00
H5	Monotone vs base	8/18/6	−0.20	−0.31	[−0.80, 0.00]	0.10	0.47

Added after the main run (exploratory). At the instructor's suggestion we added accuracy, ROC AUC, mean average precision (mAP, one-vs-rest, macro) and R^2, computed from the saved test predictions, and the six comparisons within each task, with Holm's correction over these 12. We also added Hodges–Lehmann (HL) estimates of the reduction (a pseudo-median) with exact 95% intervals [4], tests for the plan's comparison at each depth that every cascade trained (every cascade cut at depths 1–3), and a *positive control*: every method, with the main settings, on two synthetic datasets of 300 rows and six standard normal features with a pure interaction. In smoke-XOR the class is the sign of x_1x_2, with 10% of labels flipped; in smoke-pair $y = x_1 + 3x_2x_3 + \varepsilon$, $\varepsilon \sim N(0, 0.5^2)$. Here $K = 3$ products are chosen from 15 candidates. Follow-up runs on why products do not help are in Appendix B.

3 Results

3.1 Hypotheses

No hypothesis is supported (Table 1, Figure 1). Residual-XX is worse than the base cascade on 18 of 32 datasets (median −0.66%). It does no better than random products (16 wins, 16 losses) and loses less than target-chosen products: 20 wins, 9 losses ($p = 0.020$, Holm $p = 0.12$). This difference is fragile: per seed p is 0.79, 0.002 and 0.25, and its largest win, Bike Sharing, comes from target-XX adding the same products at every layer and losing 15% to the base cascade. XP-MI, non-linear and monotone columns have medians between −0.2% and 0. Their six ties are mostly regression sets where both cascades chose layer 1 in all 15 runs, so the columns, which enter from layer 2, were never used.

The intervals bound the plausible effect. The HL upper limits for H1 and H3–H5 lie between 0 and +0.36%, but HL estimates a pseudo-median and keeps the ties; with ties dropped the limits for H3–H5 rise to +0.34%, +0.58% and +0.15%, and a distribution-free interval for H1's median reaches +0.66%. With every cascade cut at layer 2, where the added columns are always used, the HL intervals for H3–H5 lie within ±0.5% and H1's lies below zero (−3.95% to −0.37%). Gains above about 0.7% are therefore unlikely on datasets like ours. No dataset reliably gains from residual-XX: by a paired test over its 15 runs it is worse on 10 datasets and better on 1. Products of every kind make the cascade slightly worse: random-XX and target-XX lose to the base cascade on 23 and 21 datasets ($p = 0.0008$ and 0.003, descriptive). The chosen XP products rarely copy the predictions (median absolute correlation 0.32; 1.6% above 0.95).

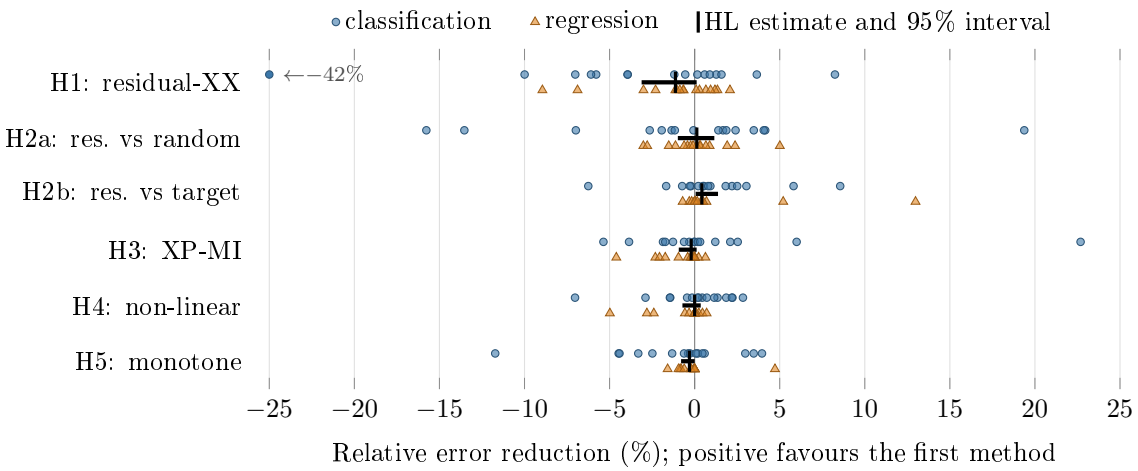


Figure 1: Relative error reduction on each dataset for the six comparisons of Table 1. One value lies outside the axis: residual-XX on Tic-Tac-Toe, −42% (error 2.3% instead of 1.4%).

Table 2: The six comparisons by task and metric: wins/losses of the first method over the 16 datasets of each task; the rest are ties. $^{*}p < 0.05$, $^{**}p < 0.01$, two-sided Wilcoxon test without correction. For the primary metrics (balanced accuracy, RMSE), Holm's correction over the 12 comparisons gives $p \geq 0.39$.

	Classification						Regression		
	Bal. acc.	Accuracy	ROC AUC	mAP	Log-loss	Macro-F1	RMSE	MAE	R^2
H1	7/9	8/8	4/12*	6/10	5/11	9/7	7/9	6/10	6/10
H2a	8/8	7/8	5/11	5/11	6/10	8/8	8/8	7/9	7/9
H2b	11/5	11/5	8/8	8/8	11/5	10/6	9/4	8/5	8/5
H3	7/8	9/6	9/7	8/8	3/13*	9/7	2/9*	3/8*	2/9*
H4	10/6	10/6	9/7	10/6	9/7	10/6	4/6	5/5	5/5
H5	7/9	7/9	2/14**	2/14*	1/15**	8/8	1/9	1/9	1/9

3.2 By task and on other metrics

Within each task no comparison is significant after Holm's correction over the 12 (smallest adjusted $p = 0.39$). Of the 54 tests in Table 2, 8 have $p < 0.05$ without correction. They come from about four distinct signals, because the metrics are strongly related, and all favour the reference. Two patterns stand out.

First, monotone columns make class probabilities worse while balanced accuracy does not change: ROC AUC is lower on 14 of 16 datasets, mAP on 14 and log-loss on 15 ($p = 0.0006$, the only test below a Bonferroni threshold over the 54). After binning, the sigmoid of a prediction column splits the training rows exactly like the column itself, so on the training rows these columns copy the predictions (held-out rows can land in other bins); without binning, so does the ReLU of a probability. The drop in ROC AUC appears with all three activations and in every sensitivity setting (2–5 wins against 10–13 losses). We suspect the extra weight the copies give the predictions at the splits, not the activation: at layer 2 the monotone variant's outputs are more confident than the base cascade's on 13 of 16 datasets. Second, XP-MI is worse in regression (RMSE: 2 wins, 9 losses, 5 ties, $p = 0.033$; MAE and R^2 move with it).

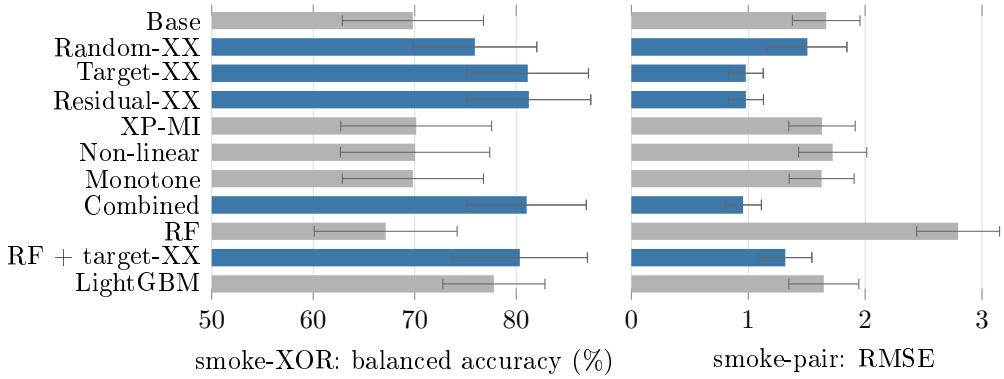


Figure 2: Positive control: mean over 15 runs, with the standard deviation as whiskers; methods that add XX products in blue. Higher balanced accuracy and lower RMSE are better; balanced accuracy starts at chance level (50%).

3.3 Positive control

Target and residual products cut the base cascade's error by about 40% (Figure 2): balanced accuracy rises from 69.8% to 81.2% on smoke-XOR, and RMSE falls from 1.67 to 0.98 on smoke-pair. Both rules put the true pair among the three products in every first-layer selection (random: 27–29%, chance: 20%). In later layers residual selection keeps it in 70% (XOR) and 94% (pair) of selections: once the predictions capture the interaction, the layer error no longer correlates with it. Random products help less, and XP-MI, non-linear and monotone columns change the error by at most 3.2%. A single forest with the target-chosen products reaches 80.4% and 1.32. The method thus finds and uses a planted interaction of exactly the form it computes, a product of mean-centred features. Section 4 discusses why this does not carry over to the real datasets.

3.4 Depth

The depth rule stops early (Table A2). The base cascade chooses layer 1 in 43% of runs (29% in classification, 57% in regression) and layer 2 in 47%. At layers 1–3 the test error is lower than the OOF error the rule sees, by a median 2.4%, 6.9% and 9.1%, so the rule undervalues depth. Probable causes, not verified, are that a test prediction averages three fold models while an OOF prediction comes from one, trained on two thirds of the rows. From layer 2 the planned reuse of folds also lets a fold model's inputs depend on its validation rows' labels, which if anything makes the OOF error optimistic. Cutting the base cascade at layer 2 would beat the chosen layer on 19 of 32 datasets ($p = 0.013$). This does not hide a gain from the added columns: at fixed depths 1–3, the comparison the plan promised, the six comparisons look the same or worse (Table A1); at layer 2, for example, residual-XX loses to the base cascade on 24 datasets ($p = 0.012$). Residual selection beats target selection at layers 2 and 3 (24/8 and 21/11).

3.5 Sensitivity checks

No setting changes the picture (Figure 3, Table A3). Against the base cascade under the same setting, residual-XX has medians from −1.04% to +0.12%, and non-linear and monotone columns stay within ±0.1%. Without binning, where an activation can add split points, monotone columns do not help with the sigmoid, ReLU or GELU (10, 10 and 7 wins against 15, 16 and 18 losses). With $2K$ products all three product variants do

slightly worse than with K (median -0.2% to -0.4% against their main-settings runs). Five OOF folds improve every tested method against its main-settings runs, the base cascade by a median 0.65% and residual-XX by 1.18% (better on 27 of 32 datasets). Only here does residual-XX beat the base cascade in median, by 0.12%, too little to matter.

3.6 Baselines and ranks

LightGBM has the best mean rank (4.03; base cascade 5.00) and beats the base cascade on 21 of 32 datasets ($p = 0.060$). The base cascade beats a single forest of 400 trees on 21 datasets ($p = 0.007$), 13 of 16 in classification. Friedman's test rejects equal ranks ($p < 0.001$), but no cascade variant differs from the base cascade by more than the critical difference (Table A4, Figure 4). In the plan's 2×2 ablation the two changes do not interact (Table A5): products cost about as much with the non-linear columns as without them (median -0.63% and -0.66%), and the non-linear columns change nothing either way. Products do not help a single forest either: RF with the target-chosen products loses to RF on 24 of 32 datasets (median -1.2%, $p = 0.057$). Per-dataset results are in Table A12.

4 Discussion and conclusions

What the results show. In the plan's setting, columns added between layers do not improve Deep Forest on these datasets. The negative result is informative: for H1, H3, H4 and H5 gains above about 0.7% are unlikely, and it holds in both tasks, on every metric, at every fixed depth and in every sensitivity check, while the median cost per run grows by 1–46%. Two findings remain useful: re-selecting products from the layer error loses less than keeping target-chosen products fixed, and products help strongly when the data hold a planted interaction of the form they compute.

Why (exploratory follow-up, Appendix B). Three findings suggest why. First, the added columns have a cost of their own. Products with their rows shuffled, the same values without information about the row, make the cascade worse on 26 of 32 datasets (median -3.0%); the unshuffled products beat them on 25 (median $+2.4\%$). The products carry information, but less than the columns cost, which fits random products hurting most and $2K$ products hurting more than K. The trees split on shuffled products mostly deep, where nodes are small. Second, on the 7 datasets where products help at layer 1 by more than 2%, the cascade's later layers gain more without them on 6. Third, a planted product explaining 6% of the target's variance is enough for products to help, but a rough measure of interaction in the real datasets does not predict where they help ($\rho = -0.13$); their interactions seem not to be of the product form. Two simpler explanations do not hold: trying the original columns at a split as often as in the base cascade does not remove the cost, and products of uncentred features do no better than the plan's (median -0.23%, 15 wins, 17 losses, below the bar set before the run), although they help on synthetic data with an interaction on the raw scale.

Limitations.

- We chose the exploratory analyses after seeing the results, and Table 2 alone holds 54 uncorrected tests. The sensitivity checks and follow-up runs use one seed and no correction.
- The relative reduction exaggerates changes when errors are close to zero (Tic-Tac-Toe: -42%); the Wilcoxon test uses ranks, which limits the weight of such a dataset.

- Parkinsons is close to unlearnable when each patient's rows stay in one fold: the mean R^2 is below zero for every method (base cascade -0.28). Without it, the tests do not change (smallest Holm $p = 0.16$).
- Columns are added only between layers; the instructor suggested interactions within a layer. Three datasets treat categories stored as numbers as numeric (Section 2); one-hot encoded, they change no conclusion (seed 0).
- In 306 of 2,640 classification runs (mostly Page Blocks, Adult and Bank Marketing) a test row gets probability 0 for its true class, so log-loss there depends on clipping. LightGBM's mean tree count reaches its 1,000-tree cap in 6 of 480 runs.

Possible continuation (not yet agreed; costs from the measured run times).

1. *Interactions within a layer* (the instructor's suggestion; its meaning is still to be agreed).
2. *Gated products:* add products at a layer only when they beat shuffled ones on the OOF predictions (a new hypothesis).
3. *Depth rule:* five OOF folds or a fixed depth improve the base cascade itself (all eight cascades with five folds: about 70 CPU-hours); this changes the cascade, not our hypotheses.
4. *Bayesian test* with a region of practical equivalence [6]: no new runs, but the result depends strongly on the agreed margin.

5 AI assistance

Claude Code (Opus 5.5) wrote the experiment code to the plan's specification, ran the experiments and analyses, and drafted this report with its figures; Codex (gpt-6-astra) reviewed the implementation plan. On 8 November, five Claude Opus and two Codex reviewers, each working alone and without changing anything, reviewed the plan, the code, the results and an earlier draft of this report; one Codex reviewer re-checked every run and number against the saved files. Claude Code proposed the follow-up runs of Appendix B from these reviews and ran them. The authors chose the analyses and runs added after the main run, and checked the results and the text.

References

[1] W. Jasiński and S. Adamiec. Non-linear and Interaction Features for Deep Forest: final plan. Ensemble Learning – Research Project, Warsaw University of Technology, 20 October 2026.

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[4] M. Hollander, D. A. Wolfe, and E. Chicken. *Nonparametric Statistical Methods*, 3rd edition, chapter 3. Wiley, 2014.

[5] J. Demšar. Statistical Comparisons of Classifiers over Multiple Data Sets. *JMLR*, 7:1–30, 2006.

[6] A. Benavoli, G. Corani, J. Demšar, and M. Zaffalon. Time for a Change: a Tutorial for Comparing Multiple Classifiers Through Bayesian Analysis. *JMLR*, 18(77):1–36, 2017.

A Additional results

Table A1: The six comparisons at the layer chosen by the depth rule (as in Table 1) and with every cascade cut at depths 1–3: wins/losses/ties and the uncorrected Wilcoxon p. Dashes: identical at layer 1 by design (XP, non-linear and monotone columns start at layer 2; residual and target selection coincide there).

	Chosen layer	Depth 1	Depth 2	Depth 3
H1	14/18/0 (0.095)	13/19/0 (0.30)	8/24/0 (0.012)	14/18/0 (0.37)
H2a	16/16/0 (0.82)	20/12/0 (0.041)	18/14/0 (0.50)	15/16/1 (0.49)
H2b	20/9/3 (0.020)	–	24/8/0 (0.0039)	21/11/0 (0.039)
H3	9/17/6 (0.27)	–	12/20/0 (0.41)	11/21/0 (0.018)
H4	14/12/6 (0.89)	–	16/16/0 (0.96)	13/19/0 (0.038)
H5	8/18/6 (0.10)	–	8/23/1 (0.15)	8/24/0 (0.0036)

Table A2: Depth rule of the base cascade. Chosen: share of runs (%) whose chosen layer is this depth, overall and by task. Gap: median over runs of the relative difference between the OOF error the rule sees and the test error (%), positive when the test error is lower. Last three columns: the base cascade cut at this depth against the base cascade at its chosen layer (median relative reduction, %; wins/losses/ties; Wilcoxon p).

Depth	Chosen (%): all, cla., reg.			Gap	Median	W/L/T	p
1	43	29	57	2.4	−1.63	6/20/6	<0.001
2	47	54	40	6.9	0.47	19/9/4	0.013
3	7	13	1	9.1	0.60	18/14/0	0.30

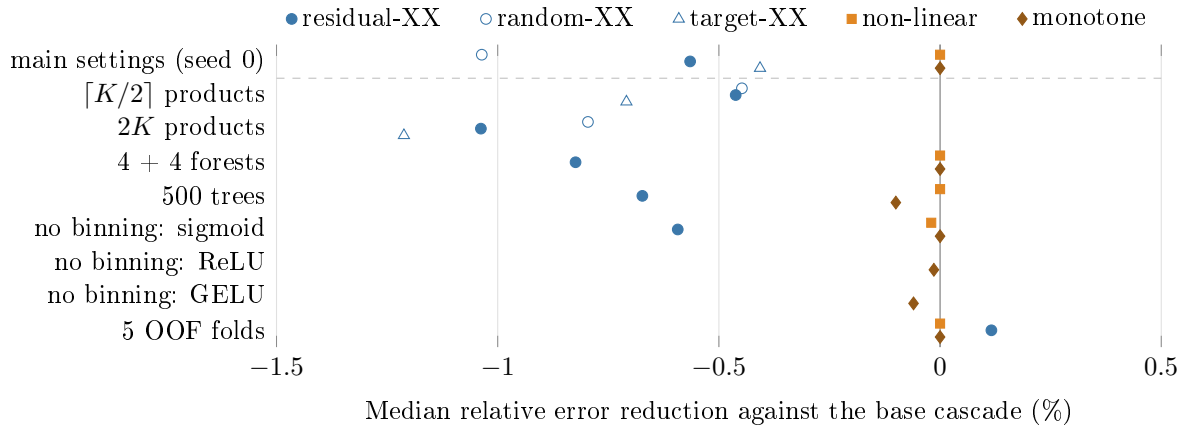


Figure 3: Sensitivity checks: each variant against the base cascade under the same setting, one seed (five runs per dataset). The top row uses the main runs of that seed. ReLU and GELU are compared with the base cascade without binning.

Table A3: Sensitivity checks, one seed. Median relative error reduction (%) and wins/losses/ties of each variant against the base cascade under the same setting (Figure 3); last column: each method against its own runs with the main settings (median, %), positive when the check lowers the error.

Setting	Method	vs base	W/L/T	vs main settings
main settings	Residual-XX	−0.56	13/18/1	–
main settings	Random-XX	−1.04	12/20/0	–
main settings	Target-XX	−0.41	13/19/0	–
main settings	Non-linear	0.00	12/13/7	–
main settings	Monotone	0.00	9/15/8	–
$\lceil K/2 \rceil$ products	Residual-XX	−0.46	11/21/0	0.27
$\lceil K/2 \rceil$ products	Random-XX	−0.45	14/18/0	0.59
$\lceil K/2 \rceil$ products	Target-XX	−0.71	11/21/0	−0.17
$2K$ products	Residual-XX	−1.04	12/20/0	−0.26
$2K$ products	Random-XX	−0.80	8/23/1	−0.16
$2K$ products	Target-XX	−1.21	10/22/0	−0.43
4 + 4 forests	Residual-XX	−0.82	11/21/0	0.01
4 + 4 forests	Non-linear	0.00	11/11/10	−0.04
4 + 4 forests	Monotone	0.00	8/13/11	0.00
4 + 4 forests	Base	–	–	0.00
500 trees	Residual-XX	−0.67	12/20/0	0.10
500 trees	Non-linear	0.00	10/14/8	0.05
500 trees	Monotone	−0.10	7/17/8	0.02
500 trees	Base	–	–	0.03
no binning, sigmoid	Residual-XX	−0.59	14/18/0	0.28
no binning, sigmoid	Non-linear	−0.02	9/17/6	0.01
no binning, sigmoid	Monotone	0.00	10/15/7	0.01
no binning, sigmoid	Base	–	–	0.11
no binning, ReLU	Monotone	−0.01	10/16/6	0.06
no binning, GELU	Monotone	−0.06	7/18/7	0.05
5 OOF folds	Residual-XX	0.12	17/15/0	1.18
5 OOF folds	Non-linear	0.00	13/15/4	1.05
5 OOF folds	Monotone	0.00	12/15/5	0.77
5 OOF folds	Base	–	–	0.65

Table A4: Mean ranks over datasets (1 = lowest error), Friedman test and Nemenyi critical difference [5], overall and by task (descriptive).

Method	All (32)	Classification (16)	Regression (16)
LightGBM	4.03	4.06	4.00
Non-linear	4.98	4.62	5.34
Base	5.00	4.97	5.03
XP-MI	5.61	5.22	6.00
Monotone	5.81	5.50	6.12
Combined	6.17	5.78	6.56
Residual-XX	6.20	5.97	6.44
RF	6.56	7.88	5.25
Random-XX	6.72	6.62	6.81
Target-XX	7.06	6.75	7.38
RF + target-XX	7.84	8.62	7.06
Friedman p	<0.001	0.0021	0.13
Critical difference	2.67	3.77	3.77

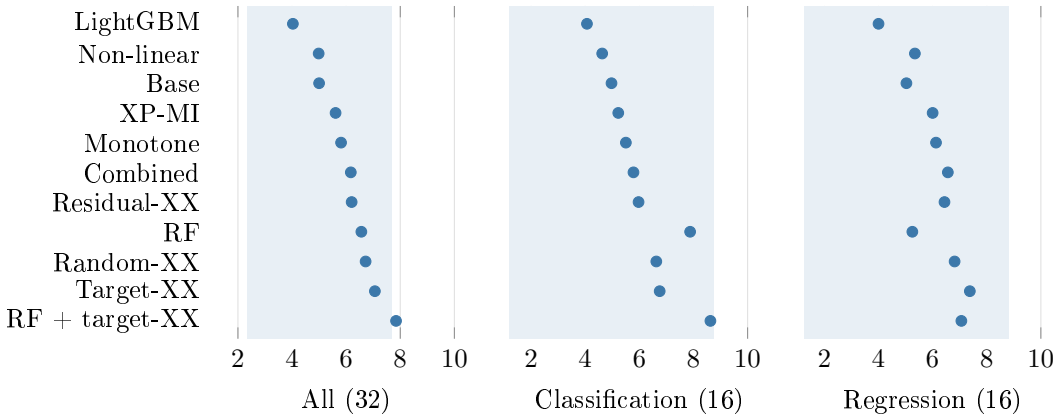


Figure 4: Rank diagram (plan §5): mean rank of each method over the datasets (1 = lowest error), from Table A4. The shaded band spans the base cascade's rank ± the Nemenyi critical difference; a method outside it differs from the base cascade (descriptive).

Table A5: The plan's 2 × 2 ablation (base cascade, non-linear, residual-XX and combined) and products in a single forest: the 32 datasets, three seeds, as in Table 1 without correction (descriptive).

Effect	Comparison	W/L/T	Median	HL	95% CI	p
Products, alone	Residual-XX vs base	14/18/0	−0.66	−1.13	[−3.12, 0.12]	0.095
Products, with non-linear	Combined vs non-linear	12/20/0	−0.63	−1.22	[−3.35, −0.09]	0.031
Non-linear, alone	Non-linear vs base	14/12/6	0.00	0.00	[−0.72, 0.36]	0.89
Non-linear, with products	Combined vs residual-XX	16/12/4	0.02	0.04	[−0.25, 0.57]	0.60
Both	Combined vs base	13/19/0	−0.58	−1.26	[−3.19, 0.01]	0.057
Products in one forest	RF + target-XX vs RF	8/24/0	−1.20	−1.37	[−3.52, 0.04]	0.057

B Follow-up runs (exploratory)

We chose these analyses after the results and the review of 8 November; they decide no hypothesis. The runs of 8 November (1,620 runs, 9 CPU-hours) and 9 November (1,205 runs, 3.4 CPU-hours, and 1.1 for the interaction measure and the split counts) use seed 0 and the main settings otherwise, with exploratory variants added to the code; with them, base and residual-XX reproduce the 320 main runs of seed 0 bit for bit. Before each run we wrote down a bar for its result.

Depth does what the products do (main runs, no new runs). Target-XX lowers the error at layer 1 by more than 2% on 7 datasets (median 3.8%, Table A6). A single forest gains about as much from the same products (Spearman $\rho = 0.79$ over the 32 datasets), so at layer 1 they add information that a forest lacks. On 6 of these 7 datasets, however, the base cascade's own later layers lower its error by more, and at the chosen layer residual-XX keeps a median gain of only 1.2%.

Table A6: Follow-up: the datasets where target-XX lowers the layer-1 error by more than 2% (main runs, three seeds; relative reduction, %). Layer 1: target-XX against the base cascade, both cut at layer 1. Single forest: RF with the target-chosen products against RF. Depth: the base cascade at its chosen layer against the base cascade at layer 1. Chosen layer: residual-XX against the base cascade, as in Table 1.

Dataset	Layer 1	Single forest	Depth	Chosen layer
Energy Efficiency	19.5	18.8	33.3	1.2
Climate Crashes	13.2	14.4	52.5	8.2
Servo	10.8	12.6	24.3	2.1
German Credit	3.8	5.3	2.7	0.9
Statlog Vehicle	3.8	2.9	10.9	−6.1
Spambase	2.6	−4.3	7.3	3.7
Page Blocks	2.6	7.8	13.4	−3.9

Information against cost. *Shuffled products* are the residual-chosen products with their rows permuted: the same values, but no information about the row. They make the cascade worse on 26 of 32 datasets (median −3.0%, $p < 0.001$; Table A7), which measures what the K extra columns cost. The unshuffled products beat them on 25 datasets (median +2.4%). The products thus carry information, but less than the columns cost, and the net is −0.6% (13 wins, 18 losses). The cost does not come from the original columns being tried less often at a split: when every original and prediction column is tried as often as in the base cascade, the shuffled products still lose 2.8% (5 wins, 27 losses), and the unshuffled products do not change (median 0.0%).

Where the cost goes. We recounted the splits of layer 1 in 96 runs (seed 0, fold 0; the test predictions equal the saved runs bit for bit). The shuffled products are half the columns and take 45% of the splits, but only 5% of the root splits and 49% of the splits below depth 10, where nodes are small (Table A8); the residual-chosen products take about half the splits at every depth. So the trees split on noise mostly where little data is left, which fits a cost of extra columns. Across datasets, however, the shuffled products' share of the splits does not track their cost (Spearman $\rho = -0.06$), against our expectation: the counts show where the cost arises, not its size.

Centring. The plan's products are built from mean-centred features. On two synthetic sets with a product of positive features on their raw scale (raw-XOR, raw-pair; Table A9),

Table A7: Follow-up comparisons on the 32 datasets, seed 0, as in Table 1 without correction. Residual: residual-XX (products of mean-centred features); raw: products of uncentred features (x/sd); shuffled: the residual-chosen products with their rows permuted; scaled: every original and prediction column is tried at a split as often as in the base cascade.

Comparison	W/L/T	Median	HL	95% CI	p
raw products vs base	15/17/0	−0.23	−0.40	[−3.06, 0.52]	0.41
raw vs centred products	17/14/1	0.31	0.27	[−1.31, 1.34]	0.65
centred products vs base	13/18/1	−0.56	−0.94	[−2.66, 0.21]	0.13
cost: shuffled products vs base	6/26/0	−3.04	−5.27	[−8.36, −2.03]	<0.001
information: products vs shuffled	25/7/0	2.39	3.09	[1.32, 8.06]	<0.001
scaled sampling: products	15/14/3	0.00	0.19	[−0.44, 2.06]	0.51
scaled sampling: shuffled	20/10/2	0.63	0.74	[−0.36, 2.31]	0.12
scaled products vs base	10/22/0	−0.66	−0.67	[−2.35, 0.82]	0.23
scaled shuffled vs base	5/27/0	−2.84	−3.55	[−7.41, −1.83]	<0.001

Table A8: Follow-up: where the trees of layer 1 split (seed 0, fold 0; median over the 32 datasets, %). Splits: share of the splits that use one of the K added columns. Samples: the same share with each split weighted by the training samples in its node. The added columns are a median 50% of the columns. Residual: residual-XX; shuffled: the same products with their rows permuted.

	Residual		Shuffled	
Node depth	Splits	Samples	Splits	Samples
All depths	49	46	45	29
Root (0)	53	53	5	5
1–2	52	49	16	11
3–5	48	49	37	24
6–10	49	47	45	37
11 and deeper	48	46	49	45

products of uncentred features (x/sd) cut the error by 19% and 5%, while the plan's products lower it by 3% on raw-XOR and raise it by 2% on raw-pair. On the real datasets uncentred products miss the bar set before the run (a positive median against the base cascade and at least 20 of 32 wins): median −0.23%, 15 wins, 17 losses.

Table A9: Follow-up synthetic controls, 1,000 rows each, mean over 15 runs: balanced accuracy (%, XOR) or RMSE (pair). Wide: the positive control's targets with 30 standard normal features, 28 (XOR) and 27 (pair) of them noise ($K = 30$). Raw: six features from $U(1, 4)$; raw-XOR's class is whether x_2x_3 exceeds its median (10% of labels flipped), and raw-pair has $y = x_2x_3 + 0.5x_1 + \varepsilon$. In the columns, raw marks products of uncentred features and T the target-chosen products.

	Base	Random	Target	Resid.	Shuffled	Target raw	Resid. raw	RF	RF + T	RF + T raw	LightGBM
wide-xor	60.3	69.7	86.9	87.0	55.4	87.9	88.0	60.2	86.7	88.0	77.5
wide-pair	2.49	2.52	1.07	1.07	2.71	1.07	1.07	2.68	1.73	1.73	1.46
raw-xor	85.0	84.9	85.5	85.5	85.4	88.0	88.0	84.4	85.0	87.8	84.9
raw-pair	0.59	0.60	0.60	0.60	0.59	0.56	0.56	0.79	0.69	0.56	0.58

Noise features. With the positive control's targets among 30 features, 27–28 of them noise (wide-XOR, wide-pair: 1,000 rows, 435 candidates, $K = 30$), residual selection still puts the true pair among the products in every first-layer selection. Residual-XX cuts the error by 67% and 57%, against 43% and 41% for LightGBM, while 30 shuffled products raise it by 11% and 8%.

How strong must an interaction be? To the 13 smaller regression datasets (all but Bike Sharing, California Housing and Superconductivity) we added two standard normal columns u_1, u_2 and replaced the target by $y + \lambda\,\mathrm{sd}(y)\,u_1u_2$, so that the product explains $\lambda^2/(1+\lambda^2)$ of the new target's variance: 0, 6, 20 or 50% for $\lambda = 0, 0.25, 0.5, 1$ (1,040 runs; groups and folds as in the main runs). Our bar: products help at a strength if residual-XX beats the base cascade in median and on at least 9 of the 13 datasets. It is met from 6% on (median +1.7%, 11 wins), and the gain grows to 10% and 25% (13 wins) at 20 and 50% (Figure 5); we had expected it to be met only at 20–50%. Residual selection puts u_1u_2 among the layer-1 products in 66%, 93% and 100% of the selections, and RF gains about as much from the target-chosen products. This is the most favourable case: an exact product of two independent columns.

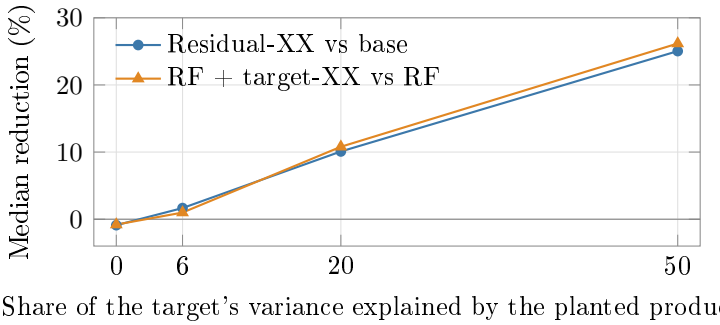


Figure 5: Follow-up: a planted product of growing strength on 13 regression datasets (seed 0); median relative error reduction over the datasets. Wins out of 13 at 0, 6, 20 and 50%: residual-XX against the base cascade 3, 11, 13, 13; RF + target-XX against RF 3, 8, 12, 13.

Interactions in the real datasets. As a rough measure we took the error reduction from LightGBM with stumps, an additive model, to LightGBM with 31-leaf trees (seed 0, five folds, early stopping on an inner split; none of the 840 fits reached the 20,000-tree cap). Planting the 6% product raises it by a median of 0.03 on the sweep datasets; 16 of the 32 real datasets score above 0.03, and Tic-Tac-Toe, Energy Efficiency and Bike Sharing above 0.5. Yet the measure does not predict where products help: its Spearman correlation with H1's per-dataset reduction is -0.13 ($p = 0.49$; the pilot gave 0.00). The measure is noisy: adding u_1, u_2 without the product moves it by up to 0.21 (Energy Efficiency).

Categories. Credit Default, Auto MPG and Energy Efficiency ran with number-coded categories (Section 2). With these one-hot encoded (165 runs), the six comparisons of seed 0 change little: no median changes sign and no Holm-adjusted p falls below 0.05 (smallest 0.27 as run, 0.31 one-hot), the bar we set before the run. One-hot encoding raises every method's error on Energy Efficiency, the base cascade's by 8% and that of RF with target-chosen products by 188%, and moves the errors on the other two datasets by at most 5%.

C Data, implementation and reproduction

Code, every setting with its source, the run summaries and the generated reports are in the repository github.com/wojtke/deep-forests, which is private until the code is released. `uv run dfx data` downloads the datasets and checks their hashes, `uv run dfx run main -workers 8` runs the main experiment and can be resumed, and `uv run dfx report main` writes the tables and figures; `sensitivity` and `positive` work the same way. Outer splits use seeds 0, 1 and 2, and every other seed is derived from the task. The per-run files with the predictions of every layer are not yet published. In the first analysis, one tie that differed only by floating-point rounding (XP-MI vs base on German Credit) counted as a win; it is now a tie, which changes H3's counts and p (0.259 to 0.269) and the base cascade's mean rank (5.02 to 5.00), but no conclusion. The figures and tables of this report are generated from the saved summaries by a script kept with its source. The follow-up runs of Appendix B use exploratory variants added to the same repository after the main runs (`dfx run` with `explore_synthetic`, `explore`, `explore_encoding` and `explore_sweep`; `scripts/interaction_strength.py` and `scripts/split_counts.py`; then `scripts/explore_report.py`). They change no setting of the main runs, but they change the code's fingerprint, so the main, sensitivity and positive runs are resumed or verified from the repository's first commit.

Table A10: Data choices the plan does not make (ours, kept after review on 25.10). Details per dataset are in the repository's data audit.

Dataset	Choice
Cardiotocography	The other expert labels of the same exam (CLASS and 10 pattern columns) dropped.
Phishing	The $\{-1, 0, 1\}$ codes treated as numeric (ordered), not categorical.
Bank Marketing	Day and month of the last contact kept as features.
Energy Efficiency	The second target, Y2 (cooling load), dropped.
Concrete Slump	The other targets, slump and flow, dropped.
Real Estate	Transaction date excluded as a timestamp.
Parkinsons	Days since recruitment (`test_time`) excluded as a timestamp.
Bike Sharing	Calendar fields (season, year, month, hour, weekday) kept as features.
Air Quality	The other reference analysers dropped as other targets; the hour excluded as a timestamp; 330 rows with every sensor missing dropped.

Table A11: Our base cascade against DF21 0.1.7 with the same five folds and preprocessing (seed 0): mean ± standard deviation over folds. Planned differences: OOF instead of out-of-bag layer predictions, balanced accuracy instead of accuracy in the depth rule.

Dataset	Metric	DF21	Ours	Layers: DF21 kept, ours chosen	
Abalone	RMSE	2.153 ± 0.054	2.147 ± 0.062	1.0	1.0
Wine Quality	RMSE	0.629 ± 0.032	0.629 ± 0.032	1.0	1.0
Spambase	Bal. acc.	0.952 ± 0.008	0.947 ± 0.010	2.4	1.8
Statlog Vehicle	Bal. acc.	0.783 ± 0.021	0.753 ± 0.035	2.0	1.6

D Per-dataset results

Table A12: Primary metric per dataset: mean ± standard deviation over 15 runs; best mean in bold. Top: balanced accuracy (%), higher is better. Bottom: RMSE in the target's units, lower is better. Every other metric is in the repository's report.

Dataset	Base	Random-XX	Target-XX	Residual-XX	XP-MI	Non-linear	Monotone	Combined	RF	RF + target-XX	LightGBM
Sonar	84.5 ± 5.0	85.5 ± 6.0	82.4 ± 4.4	82.8 ± 4.0	84.2 ± 5.1	84.8 ± 5.1	83.8 ± 4.5	83.3 ± 5.0	83.9 ± 5.2	81.7 ± 5.9	**85.7 ± 5.0**
Spambase	94.7 ± 0.8	94.7 ± 0.9	94.9 ± 0.8	94.9 ± 0.7	94.5 ± 0.9	94.7 ± 0.8	94.6 ± 0.7	**94.9 ± 0.8**	94.6 ± 0.9	94.3 ± 1.0	94.8 ± 1.1
QSAR Biodeg.	84.0 ± 3.5	84.2 ± 3.5	84.1 ± 3.6	84.0 ± 3.5	84.4 ± 3.8	84.4 ± 3.6	84.1 ± 3.5	84.1 ± 3.4	84.0 ± 3.6	83.7 ± 3.1	**84.5 ± 3.8**
Steel Plates	78.2 ± 3.9	77.6 ± 3.5	76.6 ± 2.7	77.3 ± 3.2	78.1 ± 3.5	78.4 ± 3.7	78.2 ± 3.9	77.6 ± 3.3	78.6 ± 3.1	76.6 ± 2.9	**80.2 ± 2.0**
Statlog Vehicle	76.9 ± 3.5	76.1 ± 2.6	75.9 ± 1.9	75.4 ± 2.6	77.0 ± 3.6	**77.4 ± 3.0**	76.2 ± 3.3	75.5 ± 2.1	74.6 ± 2.5	75.4 ± 2.8	77.2 ± 3.1
Statlog Heart	82.4 ± 4.6	81.9 ± 4.9	82.2 ± 4.8	82.7 ± 4.6	**83.4 ± 3.7**	82.9 ± 4.6	83.0 ± 4.1	82.6 ± 4.5	81.3 ± 4.3	80.9 ± 4.7	82.3 ± 3.1
Tic-Tac-Toe	98.6 ± 1.1	98.0 ± 1.2	97.8 ± 1.8	97.7 ± 1.9	98.9 ± 1.0	98.5 ± 1.3	98.5 ± 1.1	97.7 ± 1.9	98.3 ± 0.9	96.3 ± 1.7	**99.5 ± 0.8**
Page Blocks	85.8 ± 4.9	85.0 ± 7.3	85.1 ± 6.5	85.2 ± 6.1	86.0 ± 5.6	86.0 ± 5.3	**86.2 ± 5.4**	85.0 ± 6.3	84.3 ± 5.7	85.5 ± 5.0	84.6 ± 4.9
German Credit	66.2 ± 3.7	65.7 ± 4.0	66.6 ± 2.9	66.5 ± 3.0	66.2 ± 3.7	66.4 ± 3.8	66.4 ± 3.8	66.5 ± 2.9	65.2 ± 3.2	**67.0 ± 3.2**	66.4 ± 2.8
Cardiotocography	84.3 ± 3.2	84.3 ± 2.7	82.8 ± 3.4	83.1 ± 3.4	83.4 ± 2.5	**84.5 ± 2.9**	84.1 ± 2.7	83.1 ± 3.6	82.4 ± 3.1	80.9 ± 3.9	83.8 ± 2.6
Climate Crashes	76.2 ± 9.3	73.0 ± 9.0	76.2 ± 7.7	78.2 ± 7.5	75.9 ± 8.6	75.5 ± 9.2	75.1 ± 8.6	**78.5 ± 7.5**	56.6 ± 4.8	62.9 ± 7.8	73.1 ± 8.4
Phishing	95.3 ± 0.6	95.0 ± 0.7	95.2 ± 0.5	95.2 ± 0.5	95.3 ± 0.6	95.3 ± 0.6	95.3 ± 0.6	95.2 ± 0.5	**95.4 ± 0.6**	95.3 ± 0.6	95.4 ± 0.7
SPECTF Heart	64.1 ± 7.1	62.6 ± 5.7	59.5 ± 6.5	61.9 ± 7.3	65.0 ± 6.3	63.6 ± 6.7	**65.5 ± 8.0**	62.2 ± 7.9	60.1 ± 5.0	57.4 ± 5.2	63.4 ± 5.5
Credit Default	66.0 ± 0.8	**66.1 ± 1.1**	66.0 ± 1.0	66.1 ± 1.0	66.0 ± 1.0	66.0 ± 0.8	65.9 ± 0.8	66.0 ± 1.0	65.9 ± 0.7	65.9 ± 0.8	65.9 ± 0.9
Bank Marketing	60.3 ± 0.8	60.3 ± 0.8	60.6 ± 0.9	60.8 ± 1.0	60.2 ± 0.7	60.1 ± 0.8	60.1 ± 0.7	60.6 ± 1.0	60.7 ± 0.7	**61.1 ± 0.8**	60.8 ± 0.7
Adult	77.0 ± 0.9	76.7 ± 0.7	77.0 ± 0.8	77.1 ± 0.7	76.5 ± 1.0	76.6 ± 0.7	76.9 ± 0.8	76.9 ± 0.8	76.9 ± 0.7	76.6 ± 0.6	**78.9 ± 0.7**
Abalone	2.15 ± 0.05	2.16 ± 0.05	2.17 ± 0.05	2.17 ± 0.05	2.15 ± 0.05	2.15 ± 0.05	2.15 ± 0.05	2.17 ± 0.05	2.16 ± 0.04	2.18 ± 0.05	**2.13 ± 0.05**
Auto MPG	2.81 ± 0.34	2.78 ± 0.35	2.93 ± 0.39	2.77 ± 0.44	2.82 ± 0.34	2.81 ± 0.34	2.82 ± 0.33	2.79 ± 0.39	**2.73 ± 0.34**	2.88 ± 0.41	2.80 ± 0.30
Automobile	2,191 ± 554	2,334 ± 653	2,407 ± 667	2,406 ± 672	2,297 ± 678	2,245 ± 556	2,087 ± 472	2,436 ± 683	**2,066 ± 529**	2,183 ± 489	2,475 ± 521
Communities	0.136 ± 0.005	0.137 ± 0.005	0.137 ± 0.005	0.137 ± 0.005	0.136 ± 0.005	0.136 ± 0.005	0.136 ± 0.005	0.137 ± 0.005	0.135 ± 0.005	0.137 ± 0.005	**0.135 ± 0.005**
Servo	0.579 ± 0.211	0.597 ± 0.206	0.566 ± 0.194	0.567 ± 0.197	0.579 ± 0.212	0.609 ± 0.230	0.588 ± 0.247	0.607 ± 0.218	0.620 ± 0.161	**0.542 ± 0.152**	0.612 ± 0.173
Energy Efficiency	0.541 ± 0.151	0.545 ± 0.157	0.538 ± 0.171	0.535 ± 0.165	0.546 ± 0.151	0.544 ± 0.142	0.546 ± 0.149	0.545 ± 0.159	0.651 ± 0.161	0.529 ± 0.171	**0.406 ± 0.074**
Wine Quality	0.629 ± 0.033	0.634 ± 0.036	0.636 ± 0.037	0.635 ± 0.036	0.629 ± 0.033	0.629 ± 0.033	0.629 ± 0.033	0.635 ± 0.036	**0.629 ± 0.032**	0.634 ± 0.036	0.631 ± 0.031
Concrete Slump	3.71 ± 0.66	3.87 ± 0.62	4.00 ± 0.60	3.98 ± 0.66	3.79 ± 0.64	3.81 ± 0.65	3.72 ± 0.64	4.01 ± 0.64	3.85 ± 0.69	3.91 ± 0.59	**3.02 ± 0.41**
Real Estate	7.66 ± 1.66	**7.62 ± 1.72**	7.68 ± 1.67	7.74 ± 1.67	7.67 ± 1.73	7.62 ± 1.67	7.66 ± 1.66	7.64 ± 1.68	7.90 ± 1.65	7.98 ± 1.70	7.72 ± 1.75
Parkinsons	11.3 ± 1.6	11.3 ± 1.5	11.3 ± 1.6	11.3 ± 1.6	11.5 ± 1.7	11.4 ± 1.7	11.4 ± 1.6	11.3 ± 1.7	11.9 ± 1.3	11.9 ± 1.4	**10.9 ± 1.8**
Student Perf.	4.10 ± 0.22	4.09 ± 0.21	4.10 ± 0.23	4.10 ± 0.23	4.08 ± 0.21	4.10 ± 0.22	4.10 ± 0.22	4.09 ± 0.22	**3.98 ± 0.22**	4.08 ± 0.24	4.04 ± 0.25
Superconductivity	9.85 ± 0.37	9.82 ± 0.41	9.81 ± 0.43	9.76 ± 0.44	9.87 ± 0.39	9.78 ± 0.40	9.87 ± 0.39	9.73 ± 0.46	**9.57 ± 0.44**	9.61 ± 0.47	9.71 ± 0.45
Bike Sharing	49.6 ± 2.0	50.5 ± 2.2	58.4 ± 2.2	50.8 ± 2.0	50.8 ± 1.9	49.5 ± 2.0	50.1 ± 2.0	49.7 ± 2.2	63.0 ± 2.0	72.1 ± 2.3	**46.4 ± 1.6**
Power Plant	3.31 ± 0.08	3.37 ± 0.08	3.41 ± 0.07	3.41 ± 0.07	3.31 ± 0.08	3.31 ± 0.08	3.31 ± 0.08	3.41 ± 0.07	3.23 ± 0.08	3.33 ± 0.08	**3.15 ± 0.11**
Air Quality	0.453 ± 0.036	0.456 ± 0.035	0.452 ± 0.036	0.452 ± 0.036	0.453 ± 0.036	0.453 ± 0.036	0.453 ± 0.036	0.452 ± 0.036	**0.450 ± 0.038**	0.452 ± 0.038	0.461 ± 0.045
California Housing	0.478 ± 0.006	0.492 ± 0.008	0.479 ± 0.006	0.481 ± 0.006	0.476 ± 0.006	0.477 ± 0.006	0.481 ± 0.006	0.480 ± 0.006	0.490 ± 0.006	0.490 ± 0.005	**0.445 ± 0.008**